



Fig. 8: Top view of the robotic arm

right rotation.

Shoulder. The shoulder part of the arm is controlled by the PWM signal from pin 16 of IC1, which drives the shoulder's servo motor. Switch S4 is for 'up' movement and switch S5 for 'down' movement.

Elbow. This articulated joint is controlled by the PWM signal from pin 33 of IC1, which drives the elbow's servo motor. Switch S6 is for 'up' movement and switch S7 for 'down' movement.

Wrist. Wrist movement is controlled by the wrist's servo motor fed by the PWM signal from pin 35 of IC1. Switch S8 is for 'up' movement and

switch S9 for 'down' movement.

Grip. Grip is controlled by two servo motors. One gets the signal from pin 38, which gives 180-degree rotation to the grip. The other gets the signal from pin 29, which controls the opening and closing movement of the grip. Switch S10 is for 'anti-clockwise' movement, switch S11 for 'clockwise' movement, switch S12 for opening of the grip and switch S13 for closing of the grip.

Software

The code running on the controller is written in 'C' language and com-

plied using the Hi-Tech C compiler for 18F MCUs (evaluation mode) on MPLAB. PIC18F4431 has two modules to generate PWM signals—one is the standard capture/compare/PWM module (CCP) that generates two PWM signals and the other is the power-control PWM module that generates four PWM signals. The timers related to each PWM module are configured to generate the 20-millisecond time period and the PWM module registers are configured to generate independent PWM signals. Since PWM generation takes place in the peripheral modules, the main controller is free to load angles dynamically to control the servo joints.

There is a global variable for each joint, which holds the angle that the servo should achieve. This variable is updated as the source and destination coordinates change as the controller executes the code. A pre-calculated look-up table in the code holds the values to be loaded into the PWM module's duty-cycle registers for the corresponding angle.

EFY note. The source code of this article is available on www.efymag.com website.

This project was developed and tested at EFY Tech Center, Bengaluru

ROBO_ARM.C

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/* Servo motor PWM Programming For
PIC 18F4431. */
#include<htc.h>

void Load_Servo();
void Load_Base();
void Load_Elbow();
void Load_Wrist();
void Load_R_Grip();
void Load_L_Grip();
void Load_Default();
void manual(void);
void delay(unsigned int);

unsigned char deg_servo = 31;
unsigned char deg_base = 20;
unsigned char deg_elbow = 10;
unsigned char deg_wrist = 31;
unsigned char deg_r_grip = 31;
unsigned char deg_l_grip = 31;

/*-----Look up table for CCP PWM-----*/
unsigned const char reg_CON[4] =
{ 0x01, 0x02, 0x03, 0x04 };

unsigned const char reg_PDC[16] =
{ 0x00, 0x10, 0x20, 0x30,
  0x40, 0x50, 0x60, 0x70,
  0x80, 0x90, 0xA0, 0xB0,
  0xC0, 0xD0, 0xE0, 0xF0 };

void main()
{
/* Configures port as digital I/O */
ANSEL0 = 0x00;
ANSEL1 = 0x00;

0x07, 0x07, 0x07, 0x07,
0x08, 0x08, 0x08, 0x08,
0x09, 0x09, 0x09, 0x09,
0x0A, 0x0A, 0x0A, 0x0A,
0x0B, 0x0B, 0x0B, 0x0B,
0x0C, 0x0C, 0x0C, 0x0C,
0x0D, 0x0D, 0x0D, 0x0D,
0x0E, 0x0E, 0x0E, 0x0E,
0x0F, 0x0F, 0x0F, 0x0F,
0x10, 0x10, 0x10, 0x10,
0x11, 0x11, 0x11, 0x11,
0x12, 0x12, 0x12, 0x12,
0x13, 0x13, 0x13, 0x13 };

/* Configures the port pins as
either In/Out */
TRISA = 0x00;
TRISB = 0x00;
TRISC = 0x00;
TRISD = 0x00;
TRISE = 0x00;

/* Setting Internal Oscillator to
run at 50KHz. */
OSCCON = 0x1E;

/*-----CCP module registers-----*/
/* For generating time period 20ms */
PR2 = 0x9B;
/* Configure Timer 2 */
T2CON = 0x04;

/*---PCPWM control Module Registers---*/
PWSCON0 = 0x0F;
PTCON0 = 0x00;
PTPERL = 0x04;
PTPERH = 0x09;
PTMPL = 0x04;
PTMHH = 0x09;
PTCN1 = 0x00;
PORTA = 0xFF;

```